

Carbon Based Semi-Transparent Microelectrode Arrays

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Motivation

Microelectrode arrays (MEAs) [1] are widely used to record the cellular activity. The standard electrode materials in use are platinum, gold and titanium nitride. The entry of carbon based materials in the field of MEAs is only a decade old. Graphene and carbon nanotubes (CNTs) are extensively researched carbon based materials. Graphene application in optogenetics [2]. The impedance of the graphene based electrodes increases on reducing the size of the electrode. Here, we present a semi-transparent microelectrode array comprising CNTs as the electrode where an optimal balance between transparency, size of the electrode and impedance is achieved.

Experimental Details

A MEA substrate was fabricated utilizing indium-tin-oxide conducting paths and silicon oxide insulator with established cleanroom processes. After opening the electrodes and connection pads by reactive ion etching, a thin titanium nitride layer as a diffusion barrier followed by an ultra-thin catalyst layer of nickel is deposited. After lift-off, thermal CNT deposition (figures 1 and 2) is performed in a mixed ammonia-acetylene atmosphere [4]. The impedance measurements are performed in PBS using a Solatron SI 1260 system. The substrate transparency is measured using an UV-vis spectrometer.

Results and Discussion

Following the described process, MEAs with semi-transparent microelectrodes were fabricated. It was observed, that the thickness of the catalyst layer and the CNT height are the critical factors which determine the morphology and thus the optical transparency of the electrode layer. After process optimization, the impedance of these MEAs (electrode diameter 30 μm) was measured in the range of 100 k Ω at 1 kHz as seen in figure 3. The transparency was measured to be around 40% in the visible range, see figure 4.

In summary, the semi-transparent carbon based microelectrode array is successfully fabricated and characterized. The first cell experiments are carried out with ventricular myocytes from chick hearts and field action potentials from spontaneously beating heart cells are recorded confirming the excellent function of the carbon based electrode MEAs.

References

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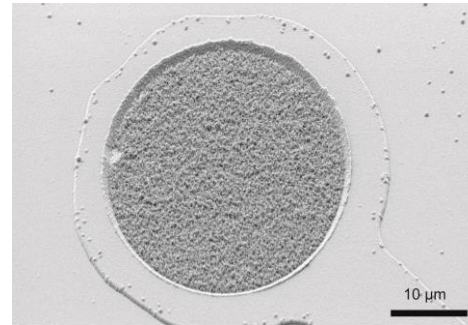


Figure 1: Scanning electron microscopy (SEM) image of the CNT microelectrode where the CNTs are grown on ITO.

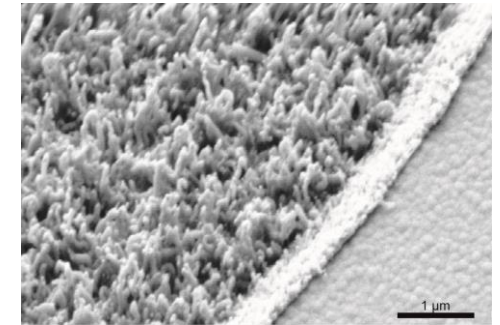


Figure 2: High magnification image SEM image of one of the electrodes with CNTs.

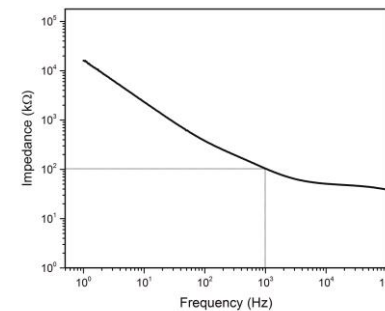


Figure 3: Impedance spectra of a typical MEA with CNTs as electrode interface. The curve depicts the linearity of impedance-frequency over a wide range.

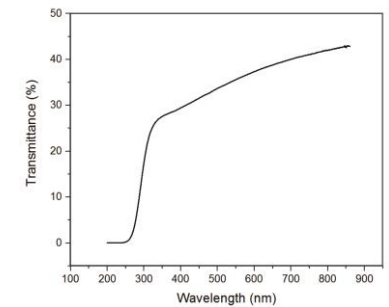


Figure 4: Transmittance curve of CNT-ITO-glass layer system resembling the MEAs.